

# Continuity



## Testing continuity

- 1 Consider the piecewise function  $f(x) = x^2 + 1$  for  $x < 2$ , and  $f(x) = 3x - 1$  for  $x \geq 2$ .
    - a Find  $\lim_{x \rightarrow 2^-} f(x)$ .
    - b Find  $\lim_{x \rightarrow 2^+} f(x)$ .
    - c Is  $f$  continuous at  $x = 2$ ? Justify.
  - 2 The piecewise function  $f(x) = kx + 1$  for  $x \leq 3$ , and  $f(x) = x^2 - 2$  for  $x > 3$ , is continuous at  $x = 3$ . Find the value of  $k$ .
- 

## Types of discontinuity and the Intermediate Value Theorem

- 3 Classify the discontinuity of each function at the given point:
    - a  $f(x) = \frac{x^2 - 1}{x - 1}$  at  $x = 1$
    - b  $f(x) = \frac{1}{x - 2}$  at  $x = 2$
    - c  $f(x) = \lfloor x \rfloor$  at  $x = 3$
  - 4 Show that  $f(x) = x^3 - 4x + 1$  has a root in the interval  $(1, 2)$ , using the Intermediate Value Theorem.
- 

## Continuity on an interval

- 5 State the largest interval(s) on which each function is continuous:
    - a  $f(x) = \frac{1}{x - 4}$
    - b  $f(x) = \sqrt{x - 3}$
    - c  $f(x) = \ln(x + 2)$
  - 6 Find the value of  $c$  that makes  $f(x) = \frac{x^2 - 9}{x - 3}$  for  $x \neq 3$ , and  $f(3) = c$ , continuous at  $x = 3$ .
- 

## Applying the Intermediate Value Theorem

- 7 Show that  $f(x) = \cos x - x$  has a root in the interval  $(0, 1)$ .

- 8 A hiker starts a trail at elevation 200 m at 8:00 AM and finishes at elevation 800 m at noon. Explain, using the IVT, why the hiker must have been at elevation exactly 500 m at some point during the hike.